

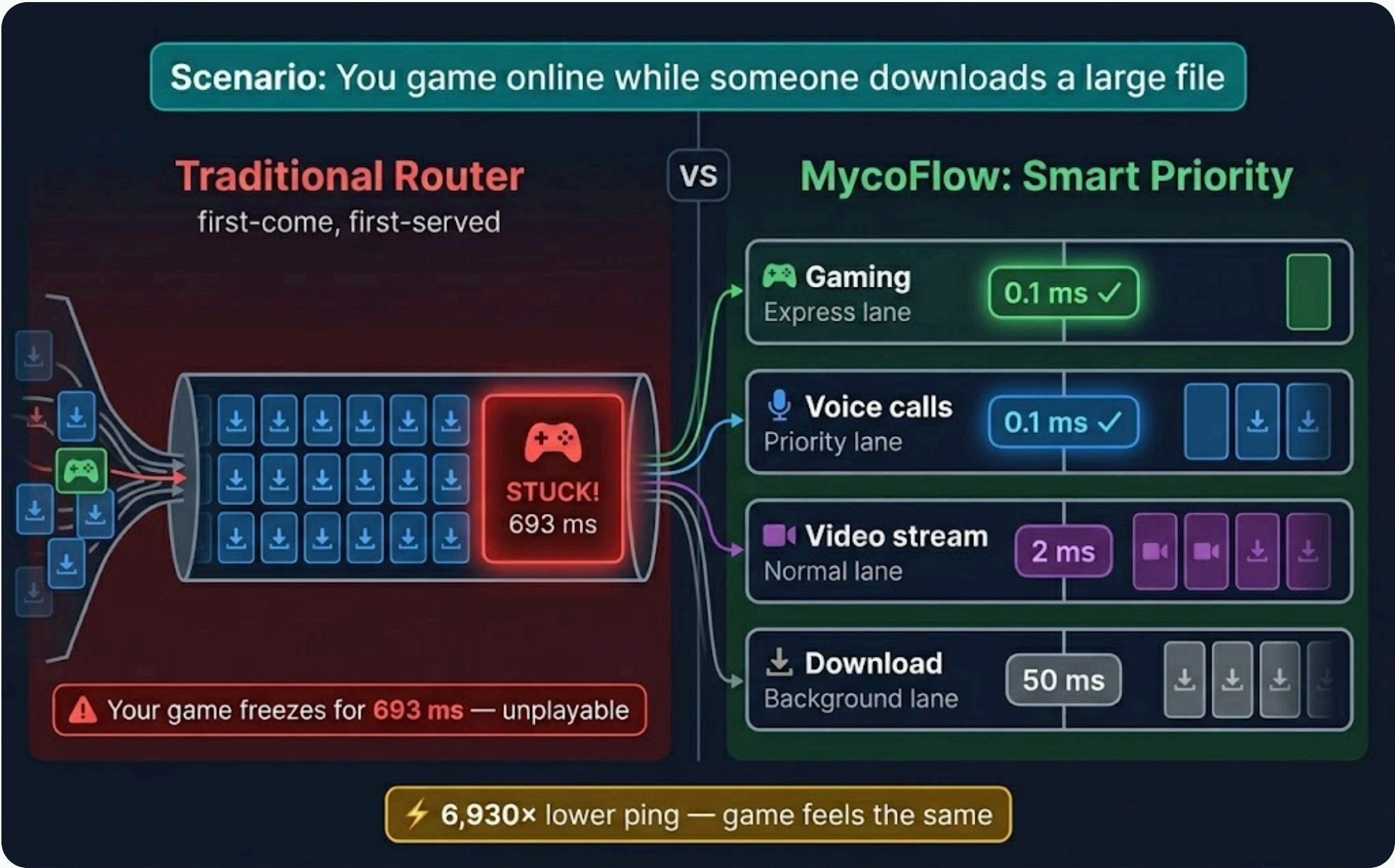
MYCOFLOW: BIOLOGICALLY INSPIRED, REFLEXIVE QOS FOR OPENWRT

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THE PROBLEM

What is Bufferbloat?



MOTIVATION

Problem

- Bufferbloat:** A single TCP flow causes 693 ms queue delay on a 20 Mbit/s WAN connection — making gaming and VoIP unusable.
- Static QoS:** Even modern AQMs like CAKE rely on DSCP tags; ISPs strip these tags, causing all traffic to fall into the same queue.
- Fixed Bandwidth:** WAN speed limit is configured once during setup; it cannot react to real-time congestion signals.

APPROACH

Solution

- 1

Biological Inspiration: Adaptive nutrient distribution strategy of fungal mycelium networks → reflexive network control
- 2

6-Class Device Persona: VOIP, GAMING, VIDEO, STREAMING, BULK, TORRENT — On-chip <1 μs decision without DPI or ML

Gaming

Streaming

VoIP

Bulk

Torrent

Unknown
- 3

12-Class Flow Taxonomy: 3-signal weighted voting combining passive DNS (0.6), target port hints (0.3), and behavioral fingerprinting (0.1)
- 4

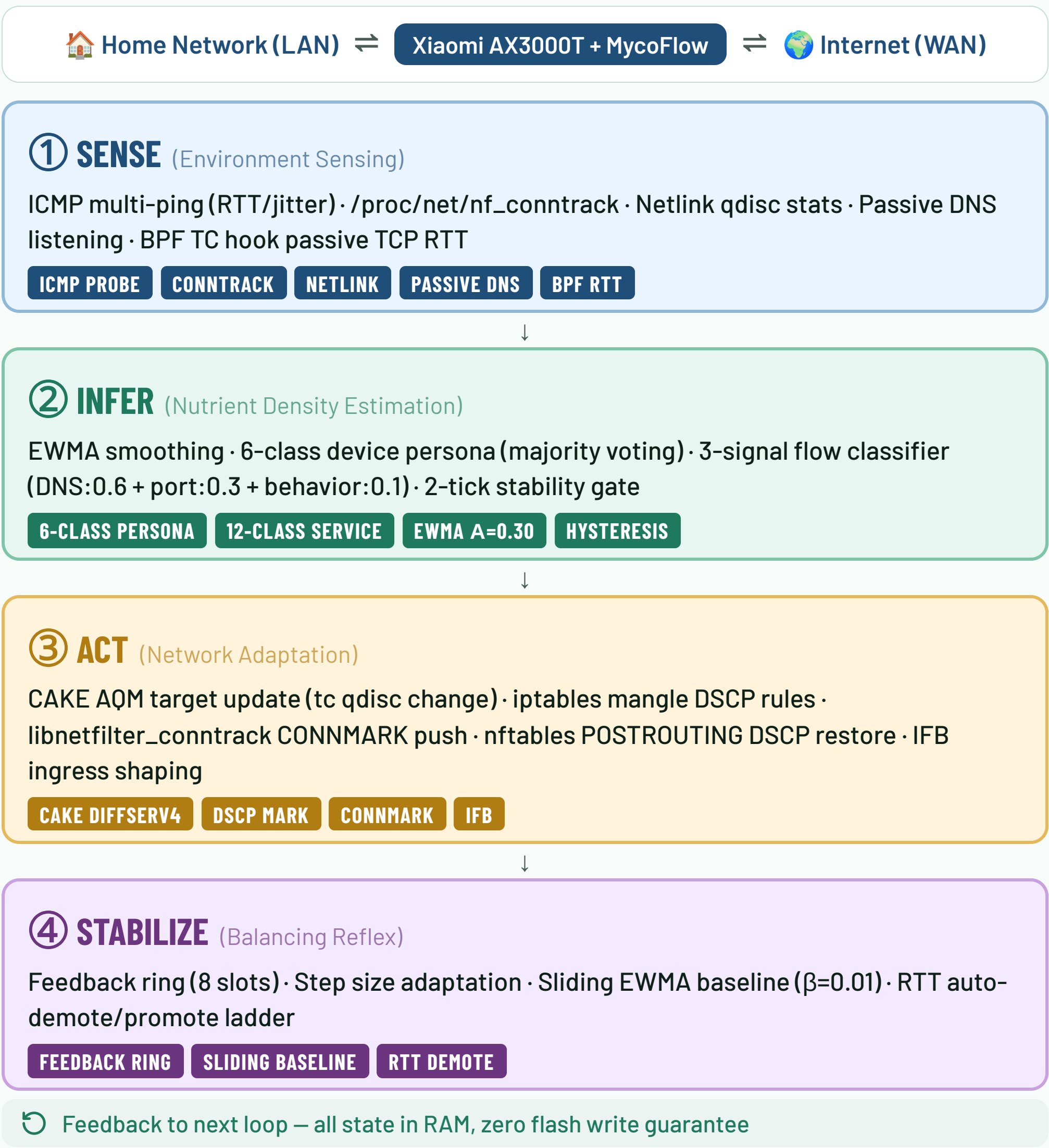
Zero Flash Writes: All state is kept in RAM — no damage to NAND flash lifespan (live tested on Xiaomi AX3000T)

COMPARISON

System	Embedded	Automatic	No ML	No DPI	Adaptive
SQM/CAKE	✓	✗	✓	✓	✗
qosify	✓	✗	✓	✗	✗
nDPI	✗	✓	✗	✗	✗
DeepFlow	✗	✓	✗	✗	✗
MycoFlow	✓	✓	✓	✓	✓

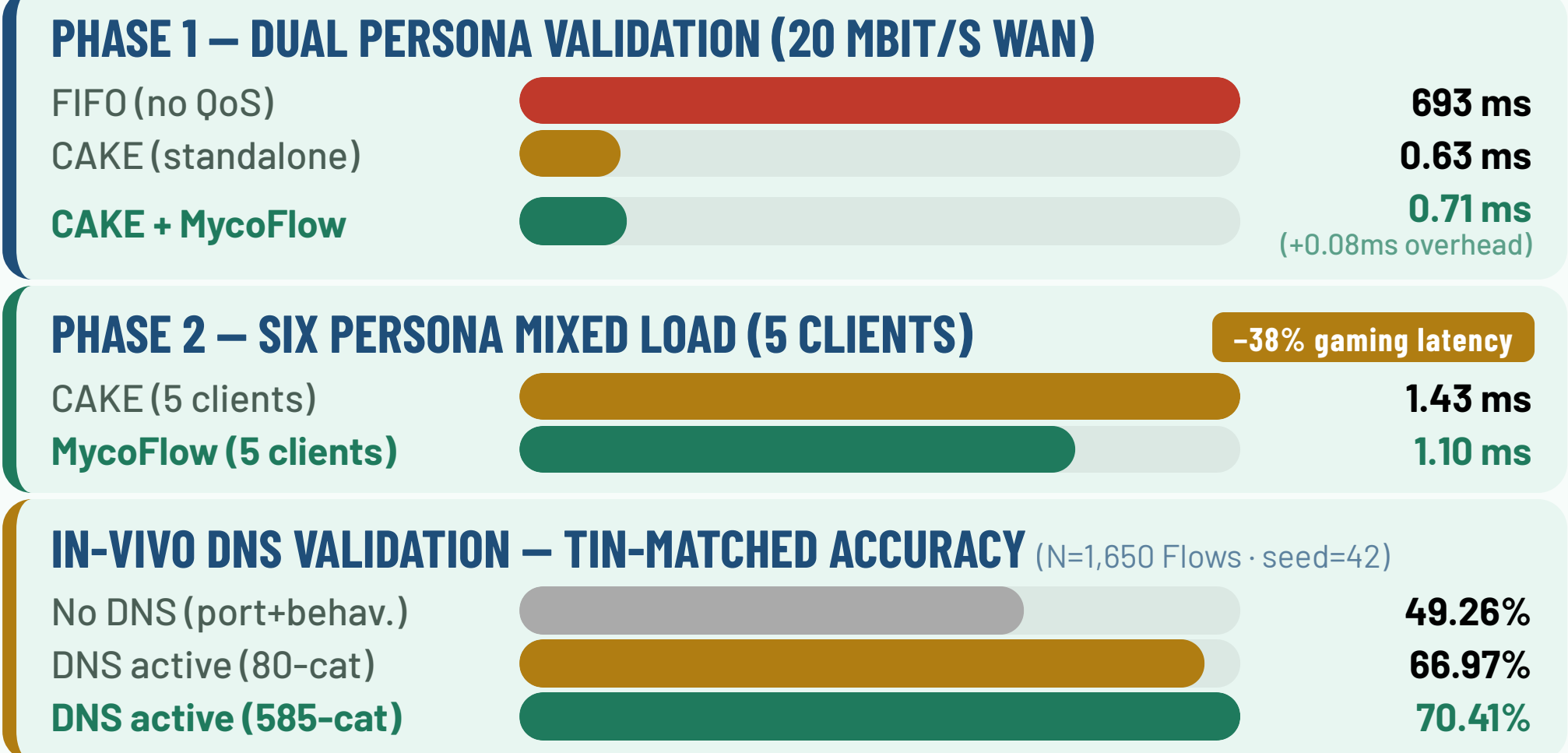
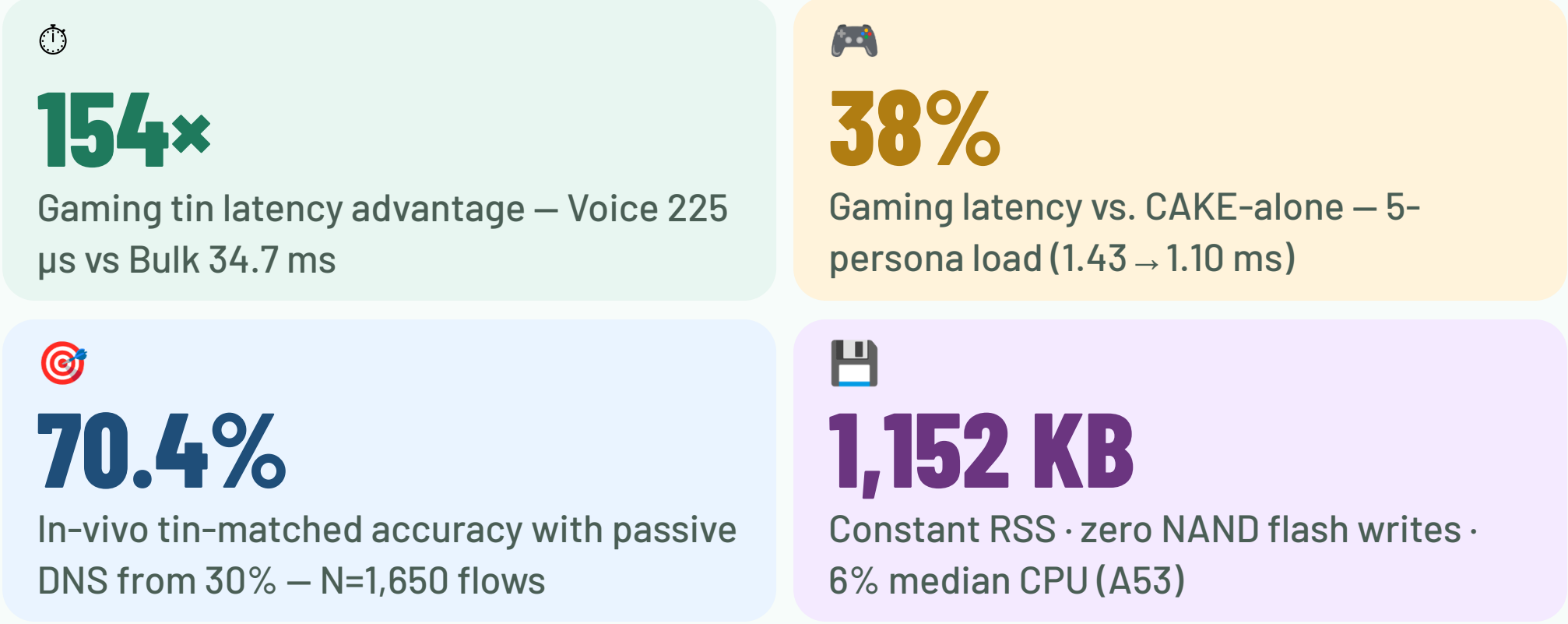
SYSTEM ARCHITECTURE

1 Hz Reflexive Control Loop



EXPERIMENTAL RESULTS

Core Metrics



Future Work: Sub-second eBPF sensing · cooperative mycelium maps · TinyML classification.

Conclusion: MycoFlow proves bio-inspired reflexive QoS on consumer CPE: 154× gaming advantage, 38% latency reduction under mixed load, 70.4% CAKE tin-matched accuracy with passive DNS — all at 1,152 KB RAM, 6% CPU, zero flash writes.

